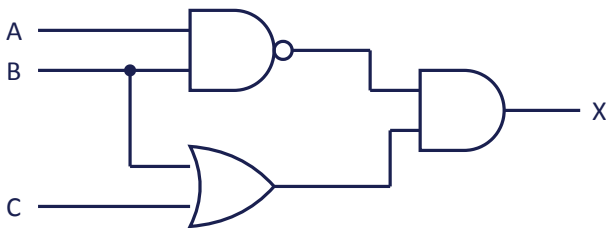


1. Complete the truth table for the logic expression $X = (\text{NOT } A) \text{ AND } (B \text{ OR } C)$. [4]

A	B	C	X
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

2. A logic circuit is shown. Write the logic expression for output X. [3]



X = _____

3. State the name of the logic gate that: [2]

(a) outputs 0 only when *both* of its inputs are 1 _____

(b) outputs 1 only when *both* of its inputs are 0 _____

4. The truth table of a logic gate is shown. State the name of the gate. [1]

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Gate: _____

5. State the number of rows in the truth table of a logic circuit with: [2]

(a) 4 inputs _____ (b) n inputs _____

6. An OR gate and an XOR gate receive the same two inputs. State the input combination for which the two gates give different outputs. [1]

A = _____ B = _____

Want to dig? Question 3 named two gates, and each of them *on its own* can build every other gate there is — they are the **universal gates**. Prove one: make NOT, then AND, then OR, using nothing but NAND. Then look up how many transistors a NAND costs against an AND, and you will see why real chips are built from the awkward gate rather than the obvious one.

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.